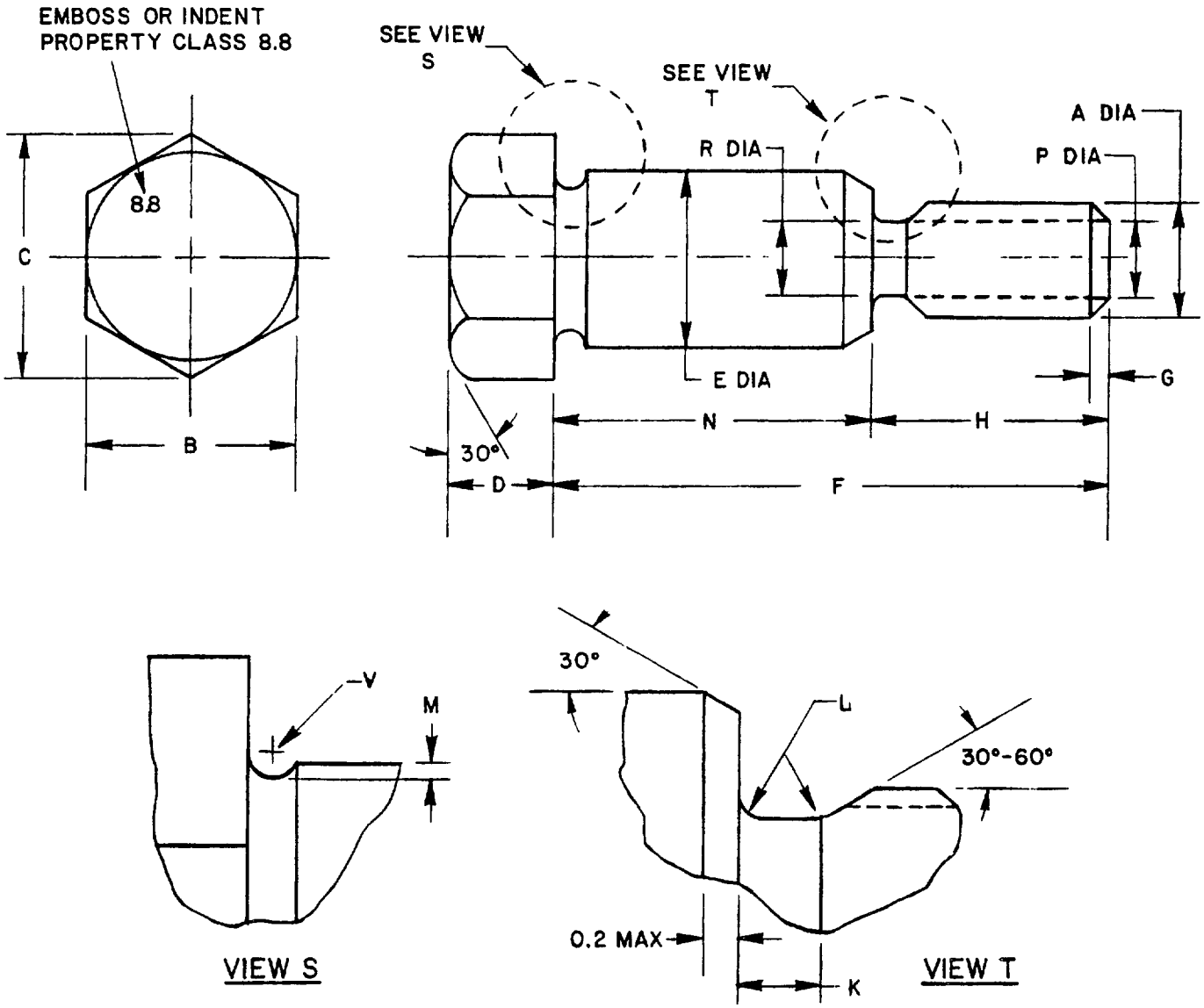
METRIC

MILITARY SPECIFICATION SHEET

BOLTS, SHOULDER, HEXAGON HEAD, LONG THREAD, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring the bolts described herein shall consist of this document and the latest issue of specification DOD-B-63475



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TABLE I Dimensions (mm)

A	Thd dia & pitch	M10X1.5	M12X1.75	M14X2	M16X2	M18X2.5	M20X2 5	M22X2.5	M24X3	M27X3	M30X3.5	M36X4
B	min	16.73	18.67	21.67	23.67	26.67	29.67	31 61	35.38	40 38	45.38	54.26
	max	17.00	19.00	22.00	24.00	27.00	30.00	32 00	36.00	41 00	46.00	55.00
C	min	18 90	21.10	24.49	26.75	30.14	33 53	35.72	39 98	45 63	51.28	61.31
D		7±0.18	8±0.18	9±0.18	10±0.18	12±0.215	13±0.215	14±0.215	15±0 215	17±0 215	19±0.26	23±0.26
E	min	11.0	13.0	15.0	17.0	19.0	21.0	23.0	25 0	28.0	32.0	38.0
	max	11.2	13.2	15.2	17.2	19 2	21 3	23.3	25.3	28 3	32.3	38.3
F		SEE TABLE II										
G	max	2.2	2 5	3.0	3 0	3.5	3 5	3 5	4 5	4 5	5.0	6.0
H(ref)	1/	19.0	22.5	24.5	27 5	30 5	31 5	35 5	—	—	—	—
	2/	21.0	24.5	26.5	29 5	32.5	33 5	37 5	40.0	43.0	47.0	54
	3/	—	—	—	—	—	—	—	—	48 0	52 0	59
K	max	1.9	2.2	2.5	2.5	3 2	3 2	3 2	3.8	3 8	4.5	5.0
L	max	1.0	1.0	1.0	1.0	1.0	1.6	1.6	1.6	1 6	1.6	2.5
M	max	0 2	0 2	0.2	0 2	0 2	0 3	0.3	0.3	0.3	0 3	0.3
N		SEE TABLE II										
P	max	7.938	9.601	11.271	13 271	15 930	16 624	19 930	19 955	24.261	25 306	30.654
R	min	7.480	9.130	10.730	12.730	14 130	16.070	18 070	19.270	22.270	24 670	29.910
	max	7.700	9.400	11 000	13.000	14.400	16.400	18 400	19.600	22 600	25 000	30.300
V	max	0.75	0 9	1.0	1.0	1.25	1.25	1.25	1.5	1 5	1 75	2 0

1/ For lengths (F) up to 50 mm incl.
2/ For lengths (F) over 50 mm to 150 mm incl.
3/ For lengths (F) above 150 mm.

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TABLE II Shoulder length - N

	M10		M12		M14		M16		M18		M20		M22		M24		M27		M30		M36	
Dim F	N	Dash no	N	Dash no	N	Dash no	N	Dash no	N	Dash no	N	Dash no	N	Dash no	N	Dash no	N	Dash no	N	Dash no	N	Dash no
30	11	-001																				
35	16	-002	12 5	-016	10.5	-032																
40	21	-003	17 5	-017	15 5	-033	12 5	-048														
45	26	-004	22 5	-018	20 5	-034	17 5	-049	14.5	-066	13 5	-083										
50	31	-005	27 5	-019	25 5	-035	22 5	-050	19 5	-067	18 5	-084	14 5	-100								
55	34	-006	30 5	-020	28 5	-036	25 5	-051	22 5	-068	21.5	-085	17 5	-101	15	-116						
60	39	-007	35 5	-021	33 5	-037	30 5	-052	27 5	-069	26 5	-086	22 5	-102	20	-117	17	-131				
65	44	-008	40 5	-022	38 5	-038	35 5	-053	32 5	-070	31 5	-087	27 5	-103	25	-118	22	-132	18	-150		
70	49	-009	45 5	-023	43 5	-039	40 5	-054	37 5	-071	36 5	-088	32 5	-104	30	-119	27	-133	23	-151	16	-168
75	54	-010	50 5	-024	48 5	-040	45 5	-055	42 5	-072	41 5	-089	37 5	-105	35	-120	32	-134	28	-152	21	-169
80	59	-011	55 5	-025	53 5	-041	50 5	-056	47 5	-073	46 5	-090	42 5	-106	40	-121	37	-135	33	-153	26	-170
85	64	-012	60 5	-026	58 5	-042	55 5	-057	52 5	-074	51 5	-091	47 5	-107	45	-122	42	-136	38	-154	31	-171
90	69	-013	65 5	-027	63 5	-043	60.5	-058	57 5	-075	56 5	-092	52 5	-108	50	-123	47	-137	43	-155	36	-172
95	74	-014	70 5	-028	68 5	-044	65 5	-059	62 5	-076	61 5	-093	57 5	-109	55	-124	52	-138	48	-156	41	-173
100	79	-015	75 5	-029	73 5	-045	70 5	-060	67.5	-077	66 5	-094	62 5	-110	60	-125	57	-139	53	-157	46	-174
110			85 5	-030	83 5	-046	80 5	-061	77 5	-078	76 5	-095	72 5	-111	70	-126	67	-140	63	-158	56	-175
120			95 5	-031	83 5	-047	90 5	-062	87 5	-079	86 5	-096	82 5	-112	80	-127	77	-141	73	-159	66	-176
130							100 5	-063	97 5	-080	96 5	-097	92 5	-113	90	-128	87	-142	83	-160	76	-177
140							110 5	-064	107 5	-081	106 5	-098	102 5	-114	100	-129	97	-143	93	-161	86	-178
150							120 5	-065	117 5	-082	116 5	-099	112 5	-115	110	-130	107	-144	103	-162	96	-179
160																	112	-145	108	-163	101	-180
170																	122	-146	118	-164	111	-181
180																	132	-147	128	-165	121	-182
190																	142	-148	138	-166	131	-183
200																	152	-149	148	-167	141	-184

NOTES

1 Dimensions Unless otherwise specified, dimensions are in millimeters (mm) and to be met after plating

2 Part number Consists of the letter "D" followed by the basic specification sheet number, a dash number taken from table II and suffix letter A or B Example D63475/1-001A Suffix "A" indicates cadmium-plated and suffix "B" zinc-plated

REQUIREMENTS

1 Material Alloy steel

2 Protective finish Cadmium plate or zinc plate (see note 2)

3 Tolerances On length (F), ±0 4 for sizes M10 thru M22 and ±0 5 for sizes M24 thru M36 On shoulder length (N) ±1 0

4 Surface hardness 20-30HRC

Custodians
Army - AR
Navy - SH
Air Force - 99

Review activities
Army - AV, MI, AT
Navy - SH
Air Force - 11
DLA-IS
NSA-NS

User activities
Army - CR, ME
Navy - AS, MC, OS

Preparing activity
Army - AR

Agent
DLA-IS

(Project 5306-0648)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

DOD-B-63475/1 Bolts, Shoulder, Hexagon Head, Long Thread, Metric

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B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

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NOTICE
OF VALIDATION

METRIC
DOD-B-63475/1
NOTICE 1
14 April 1987

MILITARY SPECIFICATION SHEET

BOLTS, SHOULDER, HEXAGON HEAD, LONG THREAD, METRIC

DOD-B-63475/1, dated 2 February 1981, has been reviewed and determined to be valid for use in acquisition.

Custodians:
Army - AR
Navy - SH
Air Force - 99

Preparing activity:
Army - AR

Review activities:
Army - AT, AV, MI
Navy - SH
Air Force - 11
DLA-IS
NSA-NS

User activities:
Army - CR, ME
Navy - AS, MC,OS

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